

Karan Mistry — Technical Lead & Full-Stack

Frappe/ERPNext Developer

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SUMMARY

- Technical Lead and Full-Stack Developer with 3+ years of hands-on experience architecting enterprise ERPs, multi-tenant SaaS products, and domain-specific business platforms on the Frappe Framework and ERPNext.
- Specialized in end-to-end system design — doctype hierarchy, workflow design, permission models, app separation strategy, and upgrade-safe deployment patterns.
- Proven track record leading developer teams, conducting code reviews, mentoring juniors, and enforcing coding standards across multiple concurrent production projects.
- Strong full-stack proficiency in Python, JavaScript, MariaDB, and Jinja, with deep production DevOps experience on Linux, Bench, Nginx, Docker, and Kubernetes.
- Hands-on experience translating complex business processes into Frappe-native solutions through BRDs, FRS, Technical Architecture Documents, ERDs, and detailed system specs.
- Active in AI-assisted development — applied context engineering, structured prompt patterns, and AI-augmented workflows across documentation, scaffolding, code review, and test generation, with strict human-in-the-loop validation at every stage.

ERPNext

- Architecting custom Frappe apps with proper module separation, doctype hierarchy, and fixture-based deployment for upgrade-safe customization across ERPNext core.
- Designing end-to-end workflows for industries including Healthcare, EPC (Engineering Procurement Construction), NBFC, Document Management, Dairy, Manufacturing, Real Estate, Trading, and HRMS.
- Building multi-tenant SaaS architectures on ERPNext with strict data isolation, tenant-specific configurations, and Frappe Cloud-equivalent auto-deployment using Docker and Kubernetes.
- Engineering production integrations with payment gateways, FHIR servers, telephony platforms (Tata Tele, Twilio, Exotel), WhatsApp Business API, Azure Blob Storage, RabbitMQ, Keycloak, and SOAP-based legacy systems.
- Performance optimization for large-scale ERPNext deployments — MariaDB query tuning, background job orchestration, Redis caching, Gunicorn worker configuration, and database replication.
- Custom reports (Script Reports, Query Reports), real-time dashboards on Frappe Insights, and Jinja-based print formats for complex billing, regulatory, and compliance scenarios.

- Workflow design including multi-stage approvals, escalations, SLA-driven notifications, and integration with Frappe Helpdesk and Frappe CRM for unified customer engagement.
- Full DevOps ownership — Linux/Ubuntu server provisioning, Bench CLI, Nginx, Supervisor, Let's Encrypt SSL, multi-site deployments, Docker, Frappe Docker, Kubernetes, and CI/CD pipelines.
- Multi-currency, multi-company ERPNext setups with localized chart of accounts, country-specific tax compliance, and consolidated cross-entity reporting.

SKILLS

- **Open Source:** ERPNext, Frappe Framework, Frappe HRMS, Frappe CRM, Frappe Helpdesk, Frappe Insights, Frappe LMS.
- **Languages:** Python, JavaScript, jQuery, SQL, HTML, CSS, Jinja2, PHP.
- **Frameworks & Libraries:** Frappe, Django, AstroJS, ReactJS, Bootstrap, Tailwind, Express.js.
- **Databases:** MariaDB, MySQL, PostgreSQL.
- **DevOps & Infrastructure:** Linux (Ubuntu), Bench CLI, Nginx, Supervisor, Let's Encrypt SSL, Docker, Frappe Docker, Kubernetes, CI/CD, multi-site deployment.
- **Integrations:** REST APIs, SOAP APIs, FHIR, Stripe, Twilio, Tata Tele, Exotel, WhatsApp Business API, RabbitMQ, Keycloak, Azure Blob Storage.
- **Architecture:** Multi-tenant SaaS, Technical Architecture Design, System Design, BRD/FRS Documentation, Pre-Sales Solutioning, Process Maturity Model (PMM).
- **Tools:** Git, GitHub, VS Code, Postman, Frappe Cloud.
- **Operating Systems:** Linux (Ubuntu), Windows.

PROJECTS

Project Name	Internal Business Operating System & Process Maturity Model (PMM)
Modules	PMS, CRM, HRMS, Sales, Delivery, Support
Description	• Architected and delivered an end-to-end internal business operating system on Frappe/ERPNext running every department from lead capture to delivery to post-go-live support. • Built multiple custom Frappe apps — Project Management System, CRM extension, HRMS extension, and a shared Fixtures app for upgrade-safe deployment across all internal apps. • Implemented a Process Maturity Model (PMM) layer governing Sales & Business Development, PMO & Delivery Excellence, and Support & Managed Services through unified weekly reviews. • Operationalized 9 live metrics in ERPNext-native reports covering lead/deal response times, demo feedback, off-track and cost-overshoot projects, project feedback, communication, and support SLAs. • Established weekly review cadence with RCA + Corrective Action + Preventive Action recorded directly in the system for accountability and continuous improvement. • Standardized 12 process templates covering Technical Architecture, Coding Standards, Git PR Workflow, Security & API Guidelines, QA, Deployment Pipeline, and Go-Live Checklist. • Built live Frappe Insights dashboards delivering real-time leadership visibility, replacing Excel-based tracking across a 60+ team members.
Role	Technical Lead & Architect
Technology	ERPNext, Frappe Framework, Python, JavaScript, MariaDB, Frappe Insights, Frappe HRMS, Frappe CRM, Frappe Helpdesk, Jinja, REST APIs, Bench CLI, Git

Project Name	Medical Telehealth SaaS
Modules	Healthcare, Accounts, CRM, HR, Finance, Purchasing
Description	- Architected a production-grade telehealth SaaS on a hybrid multi-server architecture — ERPNext for core clinical and billing logic, Express.js + PostgreSQL for mobile-facing services. - Engineered doctor-patient video consultation scheduling, session management, and real-time call orchestration. - Implemented FHIR-compliant auto-prescription generation linked to pharmacy and diagnostic lab modules for end-to-end clinical workflows. - Built an end-to-end insurance auto-billing pipeline integrated with Stripe and subscription management. - Integrated RabbitMQ as the inter-service message broker and Keycloak for centralized identity and access management. - Connected ERPNext with a Flutter mobile app and integrated DICOM imaging for medical record and image management. - Designed multi-server architecture engineered for healthcare data compliance and high availability.
Role	Full-Stack Developer & Solution Architect
Technology	ERPNext, Frappe, Python, Express.js, PostgreSQL, RabbitMQ, Keycloak, FHIR, Stripe, Twilio, Flutter, REST APIs

Project Name	Enterprise Pharmacy SaaS ERP
Modules	Pharmacy, Healthcare, Accounts, CRM
Description	- Built a multi-tenant SaaS ERP for multi-facility healthcare networks handling high-volume pharmacy operations across hundreds of branches. - Engineered a high-throughput prescription processing engine and complex multi-branch billing with consolidated invoicing. - Integrated synchronization with legacy SOAP-based medical databases for cross-system data integrity. - Customized appointment services with CMR Service, Immunization Service, and tailored questionnaires within the Patient Appointment doctype. - Integrated external billing platforms (Tebra, Office Ally) and government compliance systems (CAIR 2.0).

	• Developed React.js frontend layers for prescription operations and a patient portal for self-service medical record access. • Implemented Stripe-based subscription billing and Frappe Cloud-hosted multi-tenant architecture with strict data isolation.
Role	Frappe/ERPNext Technical Lead
Technology	ERPNext, Frappe, Python, JavaScript, MariaDB, React.js, REST APIs, SOAP APIs, Stripe, HTML, CSS, SQL

Project Name	Healthcare POS & Retail ERP
Modules	POS, Stock, Selling, Buying
Description	• Built a high-performance POS and retail ERP purpose-engineered for healthcare retail and wholesale operations. • Implemented live inventory updates with batch-wise barcode scanning for pharmaceutical-grade traceability. • Designed cashier-side expense management, wholesale-tier ledgers, and branch-wise reconciliation flows. • Built a delivery management Progressive Web App (PWA) for last-mile retail operations. • Integrated live customer display screens showing real-time cart details for transparency at the counter. • Optimized extreme keyboard-first cashier workflows and extensive operational reporting. • Integrated country-specific currency exchange APIs for live and historical rate management.
Role	Frappe/ERPNext Developer
Technology	ERPNext, Frappe, Python, Astro.js, JavaScript, MariaDB, Progressive Web App (PWA)

Project Name	Automated SaaS Deployment Platform & Country-Specific Chart of Accounts
Modules	DevOps, Accounts, Asset
Description	• Built a Frappe Cloud-equivalent multi-tenant SaaS auto-deployment system from the ground up. • Containerized Frappe applications using Frappe Docker and orchestrated deployments via Kubernetes for elastic scaling. • Designed a multi-tenant auto-provisioning pipeline for instant new client site creation. • Developed network REST APIs for HDM cash machine integration with ERPNext POS operations. • Delivered a fully customized Tanzania-specific Chart of Accounts with localized tax compliance and country-specific account structures. • Implemented customized depreciation scheduling for assets with configurable frequency and lifecycle parameters. • Built an installation framework for auto-importing custom account types, asset categories, CPA codes, and finance books based on user country selection.
Role	Frappe Developer & DevOps Engineer
Technology	ERPNext, Frappe, Python, JavaScript, Docker, Frappe Docker, Kubernetes, REST APIs

Project Name	Multi-Tenant Real Estate SaaS
Modules	CRM, Selling
Description	• Built a scalable multi-tenant real estate CRM engineered for high-concurrency property booking operations. • Designed a fully custom CRM flow within Frappe — Enquiry, Lead, Quotation, Deal Management, and post-deal handover. • Architected a property management system handling Properties, Developers, Buildings, and Units with detailed reports and dashboards. • Built dynamic lead pipelines and role-based custom dashboards configurable per tenant. • Designed personalized follow-up processes with automated reminders for inquiries and leads. • Streamlined property visit workflows with end-to-end management and tracking.

	- Utilized Frappe Custom APIs to power dynamic property and developer information on tenant-facing websites. - Engineered automated sales personnel assignment based on property developer mapping for operational efficiency.
Role	Frappe/ERPNext Developer
Technology	ERPNext, Frappe, Python, JavaScript, MariaDB, REST APIs

Project Name	Dairy Tech ERP
Modules	Manufacturing, Stock, Selling, Accounts, HR
Description	- Architected and deployed a multi-tenant Dairy ERP serving multiple dairy operations across distributed servers. - Owned the complete lifecycle — server provisioning, deployment automation, GitHub repository management, and ongoing maintenance. - Designed remote database replication for high availability across multiple server nodes. - Integrated Azure Blob Storage for scalable media and document handling at production scale. - Built mobile APIs for field-level operations including procurement, quality grading, and distribution. - Developed automated contract workflows for milk procurement and farmer settlements. - Delivered self-serve customer billing portals for distribution partners.
Role	Frappe Tech Lead & DevOps Engineer
Technology	ERPNext, Frappe, Python, JavaScript, MariaDB, REST APIs, Mobile API, Azure Blob Storage, Database Replication

Project Name	Trading Management ERP
Modules	CRM, Buying, Selling, Stock, Website
Description	• Built an end-to-end trading ERP covering the complete commercial lifecycle — procurement, inventory movement, stock valuation, and sales. • Implemented real-time operational dashboards for buying, stock, and selling cycles with predictive analytics. • Designed custom doctypes for trading-specific workflows and inventory movement tracking with branch and warehouse granularity. • Developed a dynamic website using Astro.js seamlessly connected to ERPNext via REST and custom APIs. • Integrated IndiaMart for real-time inquiry fetching with automated lead creation, duplicity checks, and approval flows. • Implemented custom follow-up flows for the sales team with team performance monitoring reports. • Built reports and analytics for future predictions and pipeline forecasting.
Role	Frappe/ERPNext & Astro.js Developer
Technology	ERPNext, Frappe, Python, Astro.js, JavaScript, MariaDB, REST APIs

Project Name	Omnichannel Social CRM (WhatsApp, Facebook, Instagram)
Modules	CRM, Selling
Description	• Built an omnichannel social CRM hub connecting ERPNext to WhatsApp Business API, Facebook Messenger, and Instagram Direct. • Centralized customer conversations across all three platforms into a single ERPNext lead/customer record. • Implemented automated lead capture from all social channels with conversation routing and session-based messaging. • Designed a user-friendly WhatsApp interface with message archives, template management, and session messaging. • Enabled bulk messaging to leads and any chosen document type with custom logs for tracking. • Built dynamic field data access from forms and static fields within ERPNext for context-aware messaging. • Implemented deep analytics tracking for engagement, response time, conversion, and sent/viewed/replied message states.

Role	Python, JavaScript & ERPNext Developer
Technology	ERPNext, Frappe, Python, JavaScript, HTML, Bootstrap, jQuery, REST APIs, WhatsApp Business API, Meta Graph API

EDUCATION

- **B.E. in Electronics and Communication Engineering**, Gujarat, India.